Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Two spheres of radii r and R carry charges q and Q respectively. When they are connected by a wire, there will be no loss of energy of the system if

(A) $q = QR$

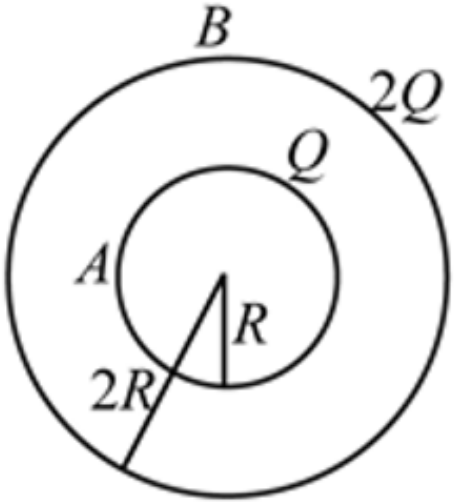
(B) $qR = Qr$

(C) $qr^2 = QR^2$

(D) $qR^2 = Q^2$

Q.2

In the figure shown, conducting shells A and B have charges Q and $2Q$ distributed uniformly over A and B.Value of $V_A - V_B$ is



(A) $\frac{Q}{4\pi\varepsilon_0 R}$

(B) $\frac{Q}{8\pi\varepsilon_0 R}$

(C) $\frac{3Q}{4\pi\varepsilon_0 R}$

(D) $\frac{3Q}{8\pi\varepsilon_0 R}$

Q.3

The electric potential at a point (x, y) is given by: $V = -Kxy$. The electric field intensity at a distance r from the origin varies as

(A) r^2

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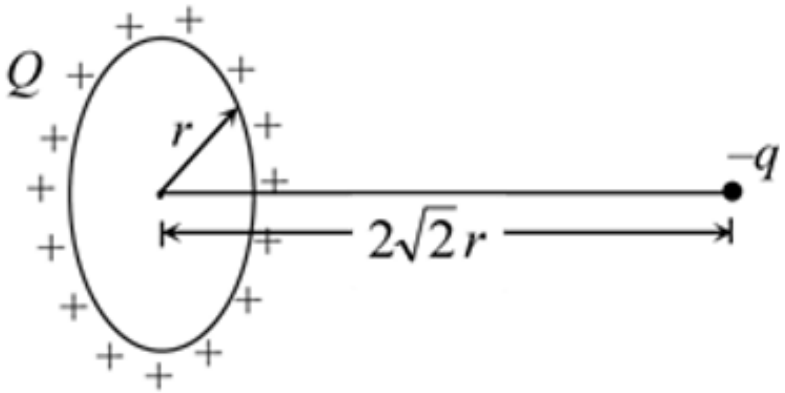
(B) r

(C) $2r$

(D) $2r^2$

Q.4

A charge $-q$ is placed at the axis of a charged ring of radius r at a distance of $2\sqrt{2}r$ as shown in figure. If ring is fixed and carrying a charge Q, the kinetic energy of charge $-q$ when it is released and reaches the centre of ring will be,



(A) $\frac{qQ}{4\pi\varepsilon_0 r}$

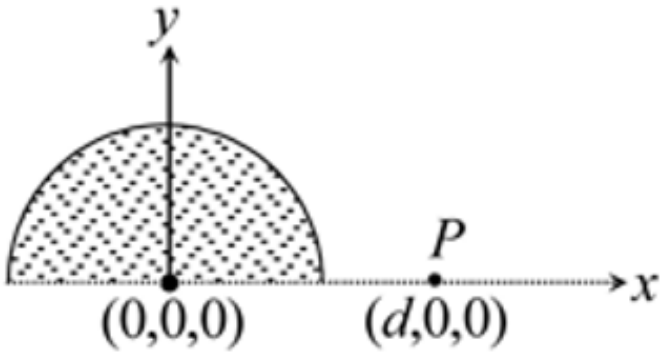
(B) $\frac{qQ}{12\pi\varepsilon_0 r}$

(C) $\frac{qQ}{6\pi\varepsilon_0 r}$

(D) $\frac{qQ}{2\pi\varepsilon_0 r}$

Q.5

A solid hemispherical uniform charged body having charge Q is kept symmetrically along the y -axis as shown in figure. The electric potential at a distance d from the origin along the x -axis at point P will be



(A) $\frac{1}{4\pi\varepsilon_0}\frac{Q}{d}$

(B) less than $\frac{1}{4\pi\varepsilon_0}\frac{Q}{d}$

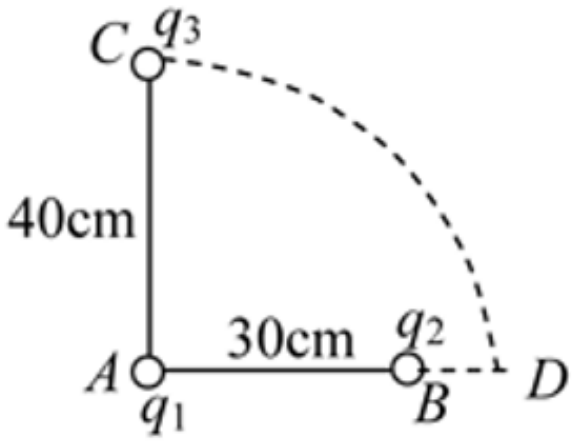
(C) more than $\frac{1}{4\pi\varepsilon_0}\frac{Q}{d}$ and less than $\frac{2}{4\pi\varepsilon_0}\frac{Q}{d}$

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(D) more than $\frac{2}{4\pi\varepsilon_0}\frac{Q}{d}$

Q.6

Two charges q_1 and q_2 are placed $30\ \text{cm}$ apart as shown. A third charge q_3 is moved along the arc of a circle of radius $40\ \text{cm}$ from C to D. The change in the potential energy of the system is $\frac{q_3}{4\pi\varepsilon_0}k$, where k is



(A) $8q_2$

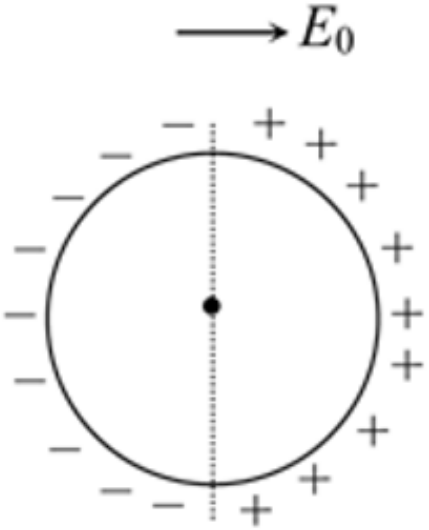
(B) $8q_1$

(C) $6q_2$

(D) $6q_1$

Q.7

Half part of ring is uniformly positively charged and other half is uniformly negatively charged. Ring is in equilibrium in uniform electric field as shown and free to rotate about an axis passing through its centre and perpendicular to plane. The equilibrium is



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(A) stable

(B) unstable

(C) neutral

(D) can be stable or unstable

Q.8

A conducting disk is of radius R is rotating with an angular velocity ω allowing the fact that electrons are the current carriers in conductor, the potential difference between the centre of the disc and edge is (mass and charge of electron is m and e and neglect gravity)

(A) $V = \frac{m\omega^2 R}{2e}$

(B) $V = \frac{m\omega^2 R^2}{2e}$

(C) $V = \frac{m0^2 R}{e}$

(D) $V = \frac{m\omega^2 R^2}{4e}$

Q.9

There is an infinite straight chain of alternating charges q and $-\mathrm{q}$. The distance between the two neighbouring charges is equal to a . Find the interaction energy of any charge with all the other charges.

(A) $-\frac{2q^2}{4\pi\varepsilon_0 a}$

(B) $\frac{2q^2 \log_e 2}{4\pi\varepsilon_0 \mathrm{a}}$

(C) $-\frac{2q^2 \log_e 2}{4\pi\varepsilon_0 a}$

(D) None of these

Q.10

Three identical particles, each possessing the mass m and charge $+q$, are placed at the corners of an equilateral triangle with side r_0. The particles are simultaneously set free and start flying apart symmetrically due to Coulomb's repulsion forces. The work performed by Coulomb's forces acting on each particle until the particles fly from one another to a very large distance is (where $\mathrm{k} = 1/4\pi\varepsilon_0$.)

(A) $\frac{3kq^2}{r_0}$

(B) $\frac{kq^2}{r_0}$

(C) $\frac{3kq^2}{2r_0}$

(D) $\frac{kq^2}{2r_0}$

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Q.11

A parallel plate capacitor having capacitance 12pF is charged by a battery to a potential difference of $10\ \text{V}$ between its plates. The charging battery is now disconnected and a porcelain slab of dielectric constant 6.5 is slipped between the plates. The work done by the capacitor on the slab is

(A) 508pJ

(B) 692pJ

(C) 560pJ

(D) 600pJ

Q.12

A hollow metal sphere of radius $5\ \text{cm}$ is charged such that the potential on its surface is $10\ \text{V}$. The potential at a distance of $2\ \text{cm}$ from the centre of the sphere is

(A) zero

(B) $10\ \text{V}$

(C) $4\ \text{V}$

(D) $10/3\ \text{V}$

Q.13

A unit charge moves on an equipotential surface from a point A to point B, then

(A) $\text{V}_\text{A} - \text{V}_\text{B} = +\text{ve}$

(B) $\text{V}_\text{A} - \text{V}_\text{B} = 0$

(C) $\text{V}_\text{A} - \text{V}_\text{B} = -\text{ve}$

(D) it is stationary

Q.14

A cube of a metal is given a positive charge Q. For this system, which of the following statements is true?

(A) Electric potential at the surface of the cube is zero

(B) Electric potential within the cube is zero

(C) Electric field is normal to the surface of the cube

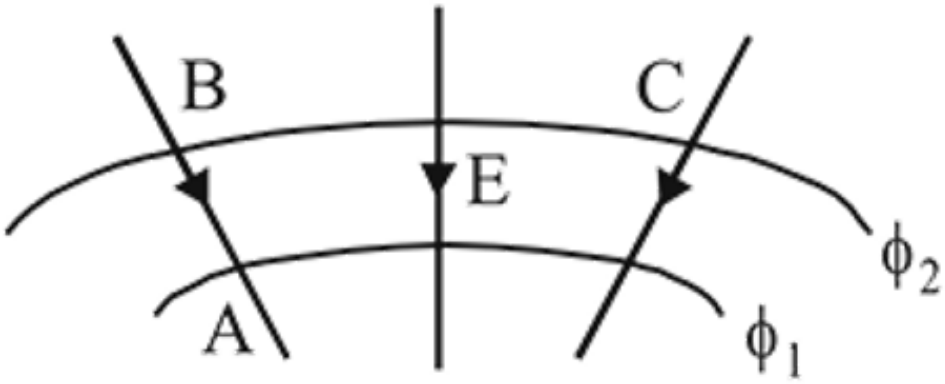
(D) Electric field varies within the cube

Q.15

In moving from A to B along an electric field line, the work done by the electric field on an electron is $6.4 \times$

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10^{-19} J. If ϕ_1 and ϕ_2 are equipotential surfaces, then the potential difference $V_C - V_A$ is



(A) -4 V

(B) 4 V

(C) zero

(D) 6.4 V

Q.16

If the electrostatic potential were given by $\phi = \phi_0 \left(x^2 + y^2 + z^2\right)$, where ϕ_0 is constant, then the charge density giving rise to the above potential would be.

(A) 0

(B) $-6\phi_0\varepsilon_0$

(C) $-2\phi_0\varepsilon_0$

(D) $-\frac{6\phi_0}{\varepsilon_0}$

Q.17

An electric charge $10^{-3}\mu$C is placed at the origin $(0, 0)$ of $X - Y$ co-ordinate system. Two points A and B are situated at $(\sqrt{2}, \sqrt{2})$ and $(2, 0)$ respectively. The potential difference between the points A and B will be [in volt]

(A) 45

(B) 9

(C) zero

(D) 2

Q.18

Two concentric spheres of radii R and r have similar charges with equal surface charge densities (σ). What is the electric potential at their common centre ?

(A) σ/ε_0

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(B) $\frac{\sigma}{\varepsilon_0}(\mathrm{R}-\mathrm{r})$

(C) $\frac{\sigma}{\varepsilon_0}(\mathrm{R}+\mathrm{r})$

(D) None of these

Q.19

Two metal pieces having a potential difference of $800\ \mathrm{V}$ are $0.02\ \mathrm{m}$ apart horizontally. A particle of mass $1.96 \times 10^{-15}\ \mathrm{kg}$ is suspended in equilibrium between the plates. If e is the elementary charge, then charge on the particle is

(A) 8

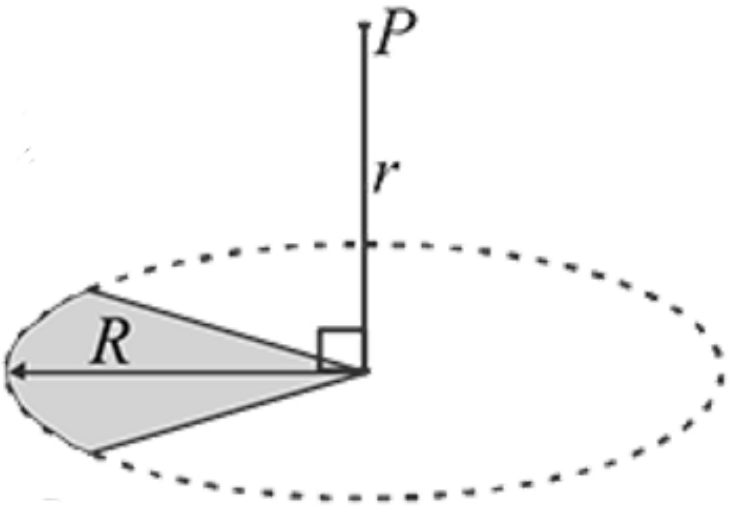
(B) 6

(C) 01

(D) 3

Q.20

A plastic disc is charged on one side with a uniform surface charge density σ and then three quadrant of the disk are removed. The remaining quadrant is shown in figure, with $\hat{V}=0$ at infinity, the potential due to the remaining quadrant at point P is



(A) $\frac{\sigma}{2\epsilon_0}\left[\left(r^2+R\right)^{1/2}-r\right]$

(B) $\frac{\sigma}{2\epsilon_0}[R-r]$

(C) $\frac{\sigma}{8\epsilon_0}\left[\left(r^2+R^2\right)^{1/2}-r\right]$

(D) None of these

Answers & Solutions

Q.1 Answer:

$qR=Qr$

Solution:

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There will be no loss of energy if the potential of the spheres is the same i.e. if

$V = \frac{q}{4\pi\varepsilon_0 r} = \frac{Q}{4\pi\varepsilon_0 R}$ **or** $\frac{q}{r} = \frac{Q}{R}$

$\therefore$ **(b)**

Q.2 Answer:

$\frac{Q}{8\pi\varepsilon_0 R}$

Solution:

$V_A = \frac{1}{4\pi\varepsilon_0}\left[\frac{Q}{R} + \frac{2Q}{2R}\right] = \frac{1}{4\pi\varepsilon_0}\frac{2Q}{R}$

$V_B = \frac{1}{4\pi\varepsilon_0}\frac{3Q}{2R}, \quad V_A - V_B = \frac{Q}{8\pi\varepsilon_0 R}$

$-$ **(b)**

Q.3 Answer:

r

Solution:

$\vec{E} = \left[\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j}\right] = K(\hat{v} + x\hat{j}) \quad \therefore \quad E = \sqrt{E_x^2 + E_y^2} = \sqrt{(Ky)^2 + (Kx)^2} = \mathrm{Kr}$

i.e., $E \propto r$

$\therefore$ **(b)**

Q.4 Answer:

$\frac{qQ}{6\pi\varepsilon_0 r}$

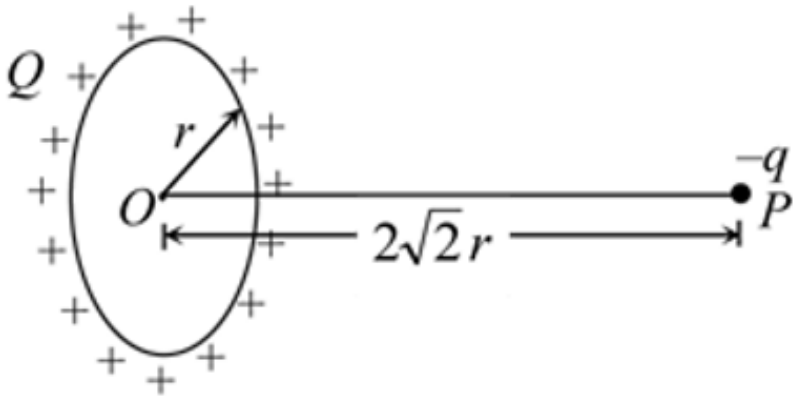
Solution:

K.E. of $-q$ **at** $O = q\,(V_O - V_P)$

$= q\left[\frac{Q}{4\pi\varepsilon_0 r} - \frac{Q}{4\pi\varepsilon_0\sqrt{r^2 + (2\sqrt{2}r)^2}}\right]$

$= \frac{2}{3}\frac{qQ}{4\pi\varepsilon_0 r} = \frac{qQ}{6\pi\varepsilon_0 r}$

$\therefore$ **(c)**



Q.5 Answer:

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$\frac{1}{4\pi\varepsilon_0}\frac{Q}{d}$

Solution:

Assuming sphere is complete then charge on it $= 2Q$

So potential at point P **due to this spherical charge** $= \frac{1}{4\pi\varepsilon_0}\frac{2Q}{d}$

Hence potential due to hemisphere $= \frac{1}{4\pi\varepsilon_0}\frac{Q}{d}$

$\therefore$ **(a)**

Q.6 Answer:

$8q_2$

Solution:

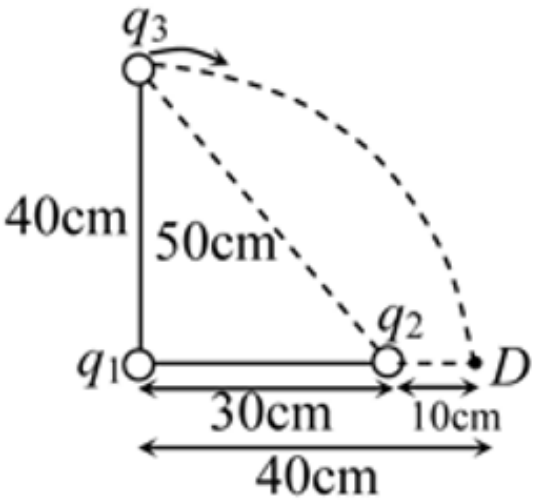
Change in potential energy $(\Delta U) = U_f - U_i$

$\Rightarrow \quad \Delta U = \frac{1}{4\pi\varepsilon_0}\left[\left(\frac{q_1q_3}{0.4} + \frac{q_2q_3}{0.1}\right) - \left(\frac{q_1q_3}{0.4} + \frac{q_2q_3}{0.5}\right)\right]$

$\Rightarrow \quad \Delta U = \frac{1}{4\pi\varepsilon_0}\left[8q_2q_3\right] = \frac{q_3}{4\pi\varepsilon_0}\left(8q_2\right)$

$\therefore \quad k = 8q_2$

$\therefore$ **(a)**



Q.7 Answer:

stable

Solution:

Assuming ring as dipole, then dipole moment $\vec{P}$ **and** $\vec{E}$ **are in same direction, so potential energy** $U = -PE$

$\therefore$

(a)

Q.8 Answer:

$V = \frac{m\omega^2R^2}{2e}$

Solution:

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Centripetal force on electron at distance r from centre of disc.

$$m\omega^2 r = eE \Rightarrow E = \frac{m\omega^2 r}{e}$$

$$\Rightarrow \quad \frac{dV}{dr} = -\frac{m\omega^2 r}{e}$$

$$\Rightarrow \quad V = -\frac{m\omega^2}{e}\int_0^R r dr = -\frac{m\omega^2 R^2}{2e}$$

$\therefore$ **(b)**

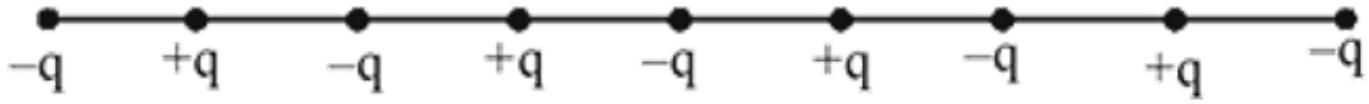
Q.9 Answer:

$-\frac{2q^2 \log_e 2}{4\pi\varepsilon_0 a}$

Solution:

$$U = 2kq^2\left[-\frac{1}{a} + \frac{1}{2a} - \frac{1}{3a} + \frac{1}{4a} + \ldots..\right]$$

$$= \frac{-\frac{2q^2}{4\pi\varepsilon_0 a}}{-q}\left[1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \ldots\right] = -\frac{2q^2 \log_e 2}{4\pi\varepsilon_0 a}$$



Q.10 Answer:

$\frac{kq^2}{r_0}$

Solution:

Since the given system is closed, the increase in KE is equal to decrease in P.E. $\Rightarrow \frac{3}{2}mv^2 = \frac{2kq^2}{r_0} - \frac{3kq^2}{r}$

$\Rightarrow v = \sqrt{\frac{2kq^2(r-r_0)}{mrr_0}}$,

v will be max when $r \to \infty \Rightarrow v_{\max} = \sqrt{\frac{2kq^2}{mr_0}}$

The work performed by the interaction force

during the variation of the system's configuration is equal to the decrease in the potential energy

$W = U_1 - U_2 = \frac{3kq^2}{r_0}$

$\therefore$ **Work done per particle** $= \frac{kq^2}{r_0}$

Q.11 Answer:

508pJ

Solution:

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Initial Energy of the capacitor, $U_i = (1/2)\mathrm{CV}^2$
$= (1/2) \times 12\mathrm{pF} \times 10 \times 10$
$= 600\mathrm{pJ}$
After the slab, the energy of the slab, $U_f = (1/2)Q^2/C'$
$Q = CV = (12\mathrm{pF})(10\ \mathrm{V}) = 120\mathrm{pC}$
$C' = kC = 6.5 \times 120 \times 10^{-12}\ \mathrm{F}$
Therefore, $U_f = \left[(1/2)\left(120 \times 10^{-2}\right)^2\right] / \left[6.5 \times 120 \times 10^{-12}\right]$
$U_f = 92\mathrm{pJ}$
$W + U_f = U_i$
$\Rightarrow W = U_i - U_f$
$= 600\mathrm{pJ} - 92\mathrm{pJ}$
$= 508\mathrm{pJ}$

Q.12 Answer:

$10\ \mathrm{V}$

Solution:

Potential at any point inside the sphere = potential at the surface of the sphere = $10\ \mathrm{V}$**.**

Q.13 Answer:

$\mathrm{V_A} - \mathrm{V_B} = 0$

Solution:

Potential should be same i.e. $\nabla_\mathrm{A} = \mathrm{V_B}$

Q.14 Answer:

Electric field is normal to the surface of the cube

Solution:

Surface of metallic cube is an equipotential surface. Therefore, electric field is normal to the surface of the cube.

Q.15 Answer:

$4\ \mathrm{V}$

Solution:

$\mathrm{W_{el.}} = \mathrm{q}\left(\mathrm{V_i} - \mathrm{V_f}\right)$
or $6.4 \times 10^{-19} = -1.6 \times 10^{-19}\left(\mathrm{V_A} - \mathrm{V_B}\right)$
or $V_A - V_B = -4V$
or $\mathrm{V_A} - \mathrm{V_C} = -4\ \mathrm{V} \quad (\because \mathrm{V_B} = \mathrm{V_C})$
or $\mathrm{V_C} - \mathrm{V_A} = 4\ \mathrm{V}$

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Q.16 Answer:

$-6\phi_0\varepsilon_0$

Solution:

$$E = -\nabla\phi = -\phi_0 2[x\hat{i} + y\hat{i} + z\hat{x}]$$
$$= \varepsilon_0 \nabla \cdot E = -2\varepsilon_0\phi_0 \nabla \cdot (x\hat{i} + y\hat{i} + z\hat{x})$$
$$n = -6\phi_0\varepsilon_0$$

Q.17 Answer:

zero

Solution:

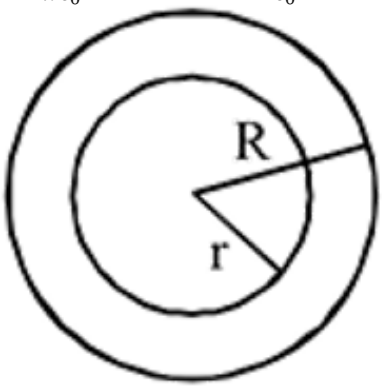
The distance of point A from origin is $\mathrm{r}_A = \sqrt{(\sqrt{2}-0)^2 + (\sqrt{2}-0)^2} = \sqrt{4} = 2$
The distance of point B **from origin is** $r_B = \sqrt{(2-0)^2 + (0-0)^2} = \sqrt{4} = 2$
Potential at A **due to charge** $\mathrm{Q} = 10^{-3}\mu\mathrm{C}$ **is** $\mathrm{V}_A = \frac{\mathrm{Q}}{4\pi\varepsilon_0 \mathrm{r_A}} = \frac{\mathrm{Q}}{8\pi\epsilon_0}$ **as** $\mathrm{r_A} = 2$
and potential at B **is** $\mathrm{V_B} = \frac{\mathrm{Q}}{4\pi\epsilon_0 \mathrm{I_B}} = \frac{\mathrm{Q}}{8\pi\varepsilon_0}$ **as** $\mathrm{r_B} = 2$
Potential difference between A **and** $\mathrm{B} = \mathrm{V_A} - \mathrm{V_B} = \frac{\mathrm{Q}}{8\pi\epsilon_0} - \frac{\mathrm{Q}}{8\pi\epsilon_0} = 0$ **volt.**

Q.18 Answer:

$\frac{\sigma}{\varepsilon_0}(\mathrm{R}+\mathrm{r})$

Solution:

Charge on the outer sphere $= q_1 = 4\pi R^2\sigma$ **Charge on the inner sphere** $= q_2 = 4\pi r^2\sigma$ $\mathrm{v} = \frac{1}{4\pi\in_0}\frac{\mathrm{q_1}}{\mathrm{R}} + \frac{1}{4\pi\epsilon_0}\frac{\mathrm{q_2}}{\mathrm{r}}$
$= \frac{1}{4\pi\epsilon_0}\left[\frac{4\pi\mathrm{R}^2\sigma}{\mathrm{R}} + \frac{4\pi\mathrm{r}^2\sigma}{\mathrm{r}}\right]$
$= \frac{4\pi\sigma}{4\pi\epsilon_0}(\mathrm{R}+\mathrm{r}) = \frac{\sigma}{\epsilon_0}(\mathrm{R}+\mathrm{r})$



Q.19 Answer:

3

Solution:

In equilibrium, $\mathrm{F} = \mathrm{qE} = (\mathrm{ne})\frac{\mathrm{V}}{\mathrm{d}} = \mathrm{mg}$
$\mathrm{n} = \frac{\mathrm{mgd}}{\mathrm{eV}} = \frac{1.96\times10^{-15}\times9.8\times0.02}{16\times10^{-19}\times800} = 3$

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Q.20 Answer:

$\frac{\sigma}{8\epsilon_0}\left[\left(r^2+R^2\right)^{1/2}-r\right]$

Solution:

The potential at P due to whole disc is $V=\frac{\sigma}{2\in_0}\left[\sqrt{R^2+r^2}-r\right]$
Now potential due to quarter disc, $V=\frac{V}{4}=\frac{\sigma}{8\in_0}\left[\sqrt{R^2+r^2}-r\right]$.

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